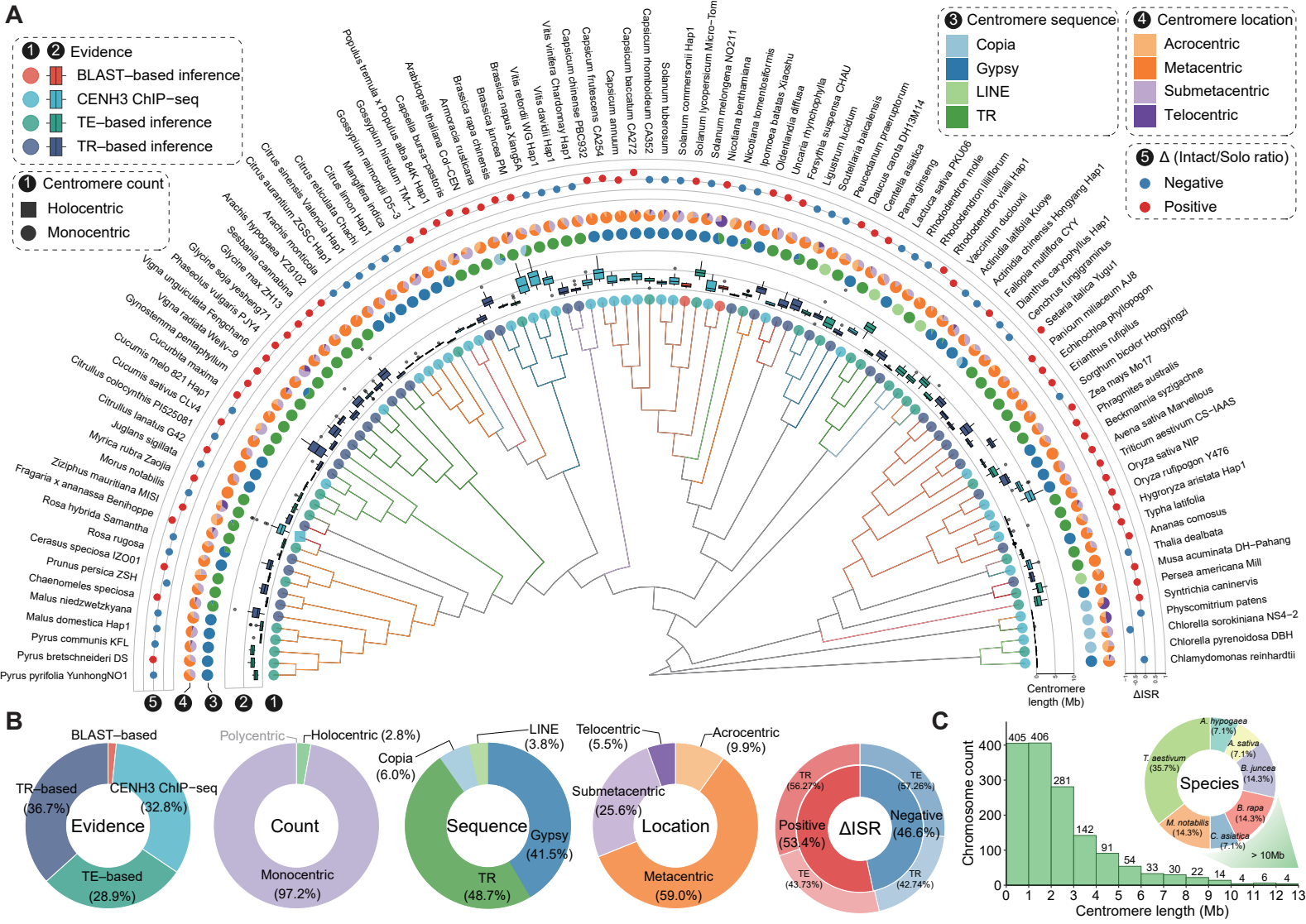
A
1 2 Evidence
BLAST–based inference
CENH3 ChIP–seq
TE–based inference
TR–based inference
1 Centromere count
Holocentric
Monocentric
3 Centromere sequence
Copia
Gypsy
LINE
TR
4 Centromere location
Acrocentric
Metacentric
Submetacentric
Telocentric
5 Δ (Intact/Solo ratio)
Negative
Positive
Populus tremula x Populus alba 84K Hap1
Gossypium hirsutum TM–1
Gossypium armourianum DS–3
Mangifera indica
Citrus limon Hap1
Citrus reticulata Chachi
Citrus sinensis Valencia Hap1
Citrus aurantium ZGSC Hap1
Arachis monticola
Arachis hypogaea YZ9102
Sesbania cannabina
Glycine max ZH13
Glycine soja yesheng71
Phaseolus vulgaris PJ.Y4
Vigna unguiculata Fengchan6
Vigna radiata Weilv–9
Gynostemma pentaphyllum
Cucurbita maxima
Cucumis melo 821 Hap1
Cucumis sativus CLv4
Citrullus colocynthis PI525081
Citrullus lanatus G42
Juglans sigillata
Myrica rubra Zaojia
Morus notabilis
Ziziphus mauritiana MISI
Fragaria x ananassa Benihoppe
Rosa hybrida Samantha
Rosa rugosa
Cerasus speciosa IZO01
Prunus persica ZSH
Chaenomeles speciosa
Malus niedzwetzkyana
Malus domestica Hap1
Pyrus communis KFL
Pyrus bretschneideri DS
Pyrus pyrifolia YunhongNO1
Arabidopsis thaliana Col–CEN
Capsella bursa–pastoris
Armoracia rusticana
Brassica rapa chinensis
Brassica juncea PM
Brassica napus XiangA
Vitis riparia WG Hap1
Vitis davidii Hap1
Vitis vinifera Chardonnay Hap1
Capsicum chinense PBC932
Capsicum frutescens CA254
Capsicum annuum
Capsicum baccatum CA272
Capsicum rhomboideum CA352
Solanum tuberosum
Solanum commersonii Hap1
Solanum lycopersicum Micro–Tom
Solanum melongena NO211
Nicotiana benthamiana
Ipomoea batatas Xiaoshu
Oldenlandia diffusa
Uncaria rhynchophylla
Forsythia suspensa CHAU
Ligustrum lucidum
Scutellaria baicalensis
Peucedanum praeruptorum
Daucus carota DH13M14
Centella asiatica
Panax ginseng
Lactuca sativa PKU06
Rhododendron molle
Rhododendron illiflorum
Rhododendron vialii Hap1
Vaccinium duclouxii
Actinidia latifolia Kuoye
Actinidia chinensis Hongyang Hap1
Fallopia multiflora CYY
Dianthus caryophyllus Hap1
Cenchrus fungigraminus
Setaria italica Yugu1
Panicum miliaceum AJ8
Echinochloa phyllopogon
Enanthus rufipilus
Sorghum bicolor Hongyingzi
Zea mays Mo17
Phragmites australis
Beckmannia syzigachne
Avena sativa Marvellous
Triticum aestivum CS–IAAS
Oryza sativa NIP
Oryza rufipogon Y476
Hygroryza aristata Hap1
Typha latifolia
Ananas comosus
Thalia dealbata
Musa acuminata DH–Pahang
Persea americana Mill
Syntrichia caninervis
Physcomitrium patens
Chlorella sorokiniana NS4–2
Chlorella pyrenoidosa DBH
Chlamydomonas reinhardtii
5 4 3 2 1
Centromere length (Mb)
ΔISR
B
BLAST–based
CENH3 ChIP–seq (32.8%)
TE–based (28.9%)
TR–based (36.7%)
Evidence
Polycentric
Holocentric (2.8%)
Count
Monocentric (97.2%)
LINE (3.8%)
Copia (6.0%)
Sequence
Gypsy (41.5%)
TR (48.7%)
Telocentric (5.5%)
Acrocentric (9.9%)
Submetacentric (25.6%)
Location
Metacentric (59.0%)
TR (56.27%)
TE (57.26%)
Positive (53.4%)
ΔISR
Negative (46.6%)
TE (43.73%)
TR (42.74%)
C
Chromosome count
405 406 281 142 91 54 33 30 22 14 4 6 4
0 1 2 3 4 5 6 7 8 9 10 11 12 13
Centromere length (Mb)
Species
A. hypogaea (7.1%)
A. sativa (7.1%)
B. juncea (14.3%)
B. rapa (14.3%)
C. asiatica (7.1%)
M. notabilis (14.3%)
T. aestivum (35.7%)
> 10Mb